

Table I: Frequency Ranges for RFID Systems

RFID frequency range	Frequency band	Coupling, and applications	Allowed field strength/transmission power
125-134.2 kHz and 140-148.5 kHz	Low frequency	Inductive coupling	72 dBμA/m
6.765-6.795 MHz	Medium frequency	Inductive coupling	42 dBμA/m
7.400-8.800 MHz	Medium frequency	Used for electronic article surveillance only	9 dBμA/m
13.553-13.567 MHz	Medium frequency	Inductive coupling, widespread usage for contactless smart-cards, smart labels and item management	42 dBμA/m
26.957-27.283 MHz	Medium frequency	Inductive coupling, special applications 42dBμA/m only	42 dBμA/m
433 MHz	UHF	Backscatter coupling, rarely used for RFID	10-100 mW
868-870 MHz	UHF	Backscatter coupling, new frequency, systems under development	500 mW, Europe only
902-928 MHz	UHF	Backscatter coupling, used for several systems	4W (spread spectrum), USA/Canada only
2.400-2.483 GHz	SHF	Backscatter coupling, used for several systems like vehicle identification	4W (spread spectrum) in USA/Canada, 500 mW in Europe
5.725-5.875 GHz	SHF	Backscatter coupling, rarely used for RFID. Potential for 802.11a and WiMAX 802.16 interference	4W in USA/Canada, 500 mW in Europe

tive coupled system as it works on Faraday's principle of magnetic induction.

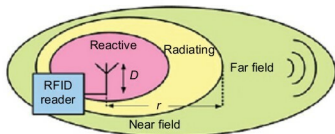


Fig. 1: Near-field and far field

The RFID reader produces an alternating magnetic field in the locality by passing a large alternating current through its coil. Now if a tag with integrated smaller coil comes under the range of this varying magnetic field, an alternating voltage induces across it. This induced voltage is rectified and coupled to a capacitor as shown in Fig. 3. The capacitor gets charged and powers the semiconductor chip for its operation.

Near-field RFID systems send the data back using load modulation

technique. As voltage is induced by inductive coupling in the tag, if there is a load a current will flow through the coil, giving rise to its own small magnetic field. This small magnetic field will oppose the reader's field. Due to this change in field, the reader's coil will have a small amount

of additional current flowing through it, which can easily be detected. The additional current in the reader's coil is actually proportional to the load across the tag's coil and hence the load modulation.

This means that if the electronics in the tag applies a variable load to its own antenna, it can actually send an encoded signal representing the ID which is detected at the RFID reader's side as variation in the current through its coil (refer Fig 3). The modulation scheme

can be chosen based on the number of bits to be transmitted, data rate and required redundancy.

The range of operation for such a system majorly depends on two things: frequency of operation and radiated energy.

The range over which we can use magnetic induction is approximately equal to $c/2\pi f$, where 'c' is the speed of light and 'f' is the frequency of operation. This means that if the frequency of operation increases, the distance over which near-field coupling can operate decreases proportionally.

The energy available for induction from the RFID reader's coil as a function of distance is another limiting factor. The magnetic field is directly proportional to $1/r^2$, where 'r' is the distance between the reader's coil and the tag. As the distance increases, there is a possibility that the magnetic field cannot induce enough energy to get the tag operative.

Inductive coupling is the most straightforward and foremost approach that was taken up for implementing an RFID system. Such wide adoption has resulted in subsequent standards. Readers can refer to ISO 15693 and 14443 for complete understanding.

Such a system works well for simple applications but when the application requires discrimination between multiple tags in the same common area for a fixed read time, each tag is required to have a higher data rate and thus higher operating frequency, which makes the design complicated. Such complication led to RFID system designs based on far-field communication.

Far-field RFID system. RFID tags based on far-field emissions capture

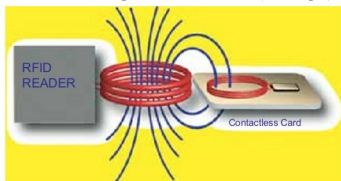


Fig. 2: Inductive coupled system